

SUJAN KS

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Portfolio : <https://port-folio-sujan.vercel.app/>



EDUCATION

N.M.A.M. Institute of Technology Nitte,India

Artificial Intelligence and Machine Learning

CGPA : 8.56

2023 — 2027

WORK EXPERIENCE

AI/ML Intern — ISIRI Technologies Pvt. Ltd. [GitHub](#)

Jun 2026 – Aug 2026

- Developed an end-to-end invoice processing system using FastAPI to extract structured data from PDF, image, Excel, and CSV invoices.
- Built a hybrid extraction pipeline combining EasyOCR, NLP preprocessing, Regular Expressions, rule-based extraction, and LLM fallback for handling incomplete or inconsistent invoice data.
- Designed REST APIs and standardized JSON data models using Pydantic to process invoice headers, supplier/buyer details, line items, pricing, taxes, batch, expiry, and totals.
- Implemented validation and quality checks for required fields, mathematical consistency, invoice totals, and extracted line-item data.
- Containerized the application using Docker and deployed the FastAPI backend on Render, integrating the components into an end-to-end document processing workflow.

Machine Learning Intern — FlyRank.ai [GitHub](#)

july 2026 - August 2026

- Built machine learning baselines and content-refresh opportunity scoring using search-performance and historical click signals
- Performed data validation, feature analysis, leakage detection, and signal auditing on large-scale content-performance data
- Developed a transparent ranking and review pipeline, achieving 0.680 Precision@50 during evaluation
- Generated ranked review queues and conducted Top-20 human/error analysis to identify model errors and support content-refresh decisions

PROJECTS

RAG-Powered Portfolio AI Assistant – Full-Stack AI Application [GitHub](#)

- Built a full-stack AI assistant using Next.js, React, and TypeScript with server-side API routes for interactive portfolio-based question answering
- Developed a RAG pipeline using Markdown knowledge bases, BAAI/bge-m3 embedding model, and Supabase pgvector for context-aware retrieval
- Implemented a 3-tier AI architecture combining local RAG, cloud LLM fallback, and rule-based fallback for reliable responses
- Integrated English/Japanese support, conversational context, AI-driven navigation, and project highlighting into a responsive chat interface

GeoSentinel – Autonomous Vision-Based Landslide Monitoring System [GitHub](#)

- Built a computer vision application for landslide detection from drone imagery, supporting real-time monitoring.
- Implemented YOLOv8 segmentation for image and video analysis, achieving 0.731 mAP@0.50.
- Integrated risk classification, heat-map visualization, and automated alerts for early hazard detection.

SmartQ Generator – Multilingual NLP & TTS System using T5 [GitHub](#)

- Built a multilingual question-generation application using T5 Transformers for generating questions from text and speech inputs.
- Integrated Speech-to-Text, language translation, and Text-to-Speech components to support multilingual interaction.
- Developed features for real-time question generation, editing, and PDF/TXT export.

Technical Skills

- **Programming:** Python, C++, Java, SQL
- **Software Engineering:** Data Structures, Object-Oriented Programming, DBMS, APIs, Back-end Development
- **Backend & Web:** FastAPI, React, TypeScript, JavaScript, Streamlit, Tailwind CSS
- **Databases:** MySQL, PostgreSQL, Supabase, pgvector
- **Cloud & DevOps:** Docker, Cloud Deployment, Git, GitHub
- **Systems & Tools:** Linux, Ubuntu, ROS2, VS Code, Jupyter Notebook, Google Colab
- **AI/ML:** Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, RAG
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, Transformers, YOLOv8, OpenCV, NumPy, Pandas

LANGUAGE PROFICIENCY

Japanese: Basic Proficiency — JLPT N5 Preparation; Guided by Professor Harikrishnan Bhat and Yasako Sato Sensei

Kannada: Native

Hindi : Professional working proficiency

EXTRACURRICULAR ACTIVITIES

- Rajya Puraskar Award – Bharat Scouts and Guides, recognizing leadership, discipline, and community service activities.
- Conducted technology awareness sessions for government school students, introducing them to basic technology and digital tools.

CERTIFICATIONS

- AI for Agriculture — Intel & Swayam Plus
- Model Context Protocol (MCP) — Anthropic
- Fundamentals of Artificial Intelligence — IBM
- Machine Learning with Python — EdiGlobe